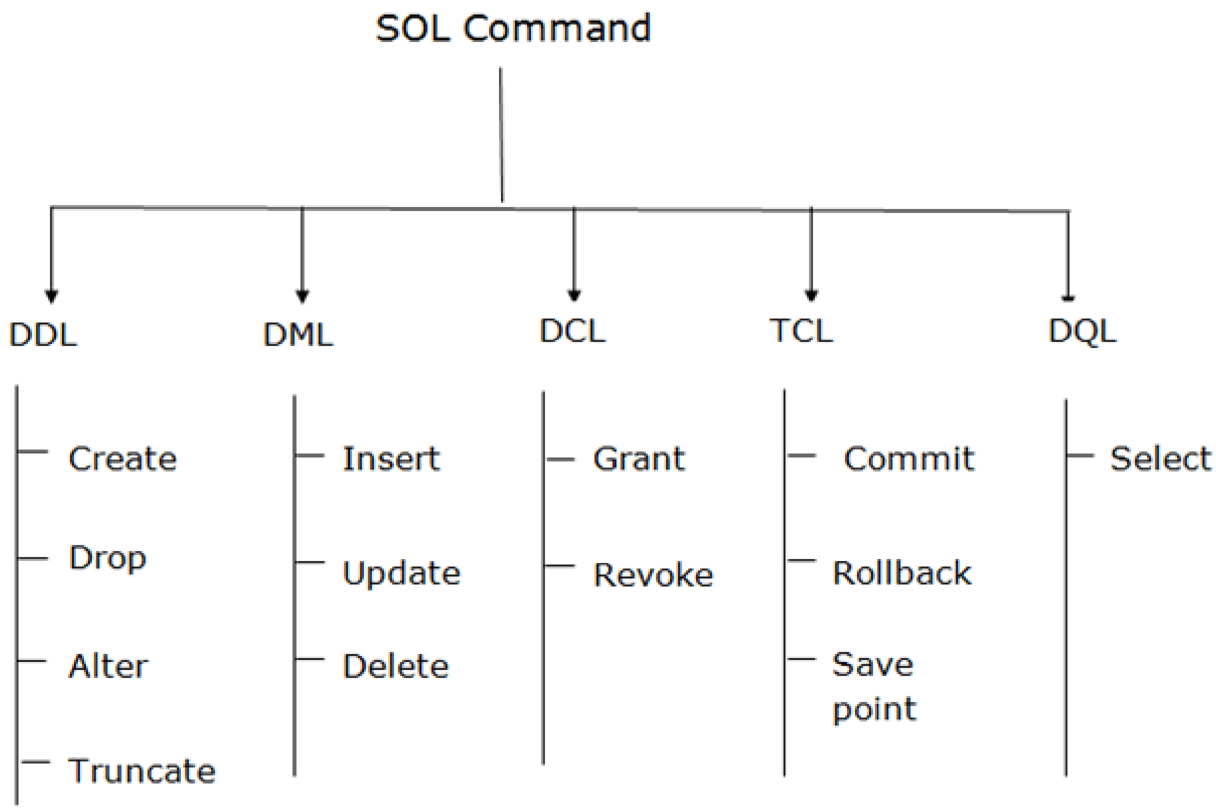
SQL Commands (DDL, DML,DQL) & SQL Constraints

SQL Commands

In SQL (Structured Query Language), there are five main types of commands:

- DDL : Data Definition Language
- DML : Data Manipulation Language
- DQL : Data Query Language
- DCL : Data Control Language
- TCL : Transaction Control Language



Data Definition Language

- Data Definition Language consists of the SQL commands that can be used to define the Database schema.
- It simply deals with descriptions of the database schema and is used to create and modify the

structure of database objects in the database.
● CREATE, ALTER, TRUNCATE and DROP are DDL commands.

1. create

```
CREATE TABLE table_name
(
column_Name1 data_type ( size of the column ) ,
column_Name2 data_type ( size of the column) ,
column_Name3 data_type ( size of the column) ,
...
column_NameN data_type ( size of the column )
) ;
```

Example:

```
CREATE DATABASE india;
USE india;
CREATE TABLE kerala
(
Roll_No int ,
First_Name Varchar (20) ,
Last_Name Varchar (20) ,
Age Int ,
Marks Int    );
```

Example 2

ADD A PRIMARY KEY:

ALTER TABLE table_name ADD PRIMARY KEY

(column_name);

REMOVE A PRIMARY KEY:

ALTER TABLE table_name DROP PRIMARY KEY;

Rules for naming tables and columns

● The name may contain letters (A-Z, a-z), digits (0-9), under score (_) and dollar ($) symbol.

● The name must contain at least one character. (Names with only digits are invalid).

● The name must not contain white spaces, special symbols.

● The name must not be an SQL keyword.

● The name should not duplicate with the names of other tables in the same database and with
other columns in the same table.

2. Drop

DROP is a DDL command used to delete/remove the database objects from the SQL database. We can easily remove the entire table, view, or index from the database using this DDL command.

DROP DATABASE Database_Name;

DROP TABLE Table_Name;

3. ALTER Command

ALTER is a DDL command which changes or modifies the existing structure of the database, and it also changes the schema of database objects.

We can also add and drop constraints of the table using the ALTER command.

ALTER TABLE - ADD Column

To add a column in a table, use the following syntax:

```
ALTER TABLE table_name
ADD column_name datatype;
```

The following SQL adds an "Email" column to the "Customers" table:

```
ALTER TABLE Customers
ADD Email varchar(255);
```

FIRST and AFTER clauses are used in the ALTER TABLE statement to specify the position where
a new column should be added within a table. These clauses are used when you want to specify
whether the new column should be the first column or be positioned before an existing column.
Add a new column at the beginning

```
ALTER TABLE table_name ADD COLUMN new_column_name data_type FIRST;
```

Add a new column after existing column

```
ALTER TABLE table_name ADD COLUMN new_column_name data_type AFTER
existing_column_name;
```

ALTER TABLE - DROP COLUMN

To delete a column in a table, use the following syntax (notice that some database systems don't allow deleting a column):

```
ALTER TABLE table_name

DROP COLUMN column_name;
```

The following SQL deletes the "Email" column from the "Customers" table:

```
ALTER TABLE Customers

DROP COLUMN Email;
```

ALTER TABLE - RENAME COLUMN

To rename a column in a table, use the following syntax:

```
ALTER TABLE table_name

RENAME COLUMN old_name to new_name;
```

ADD A PRIMARY KEY:

```
ALTER TABLE table_name ADD PRIMARY KEY

(column_name);
```

REMOVE A PRIMARY KEY:

```
ALTER TABLE table_name DROP PRIMARY KEY;
```

4. TRUNCATE

The TRUNCATE TABLE statement is used to delete all rows from a table quickly and efficiently, without logging individual row deletions. Unlike the DELETE statement, which removes rows one by one and generates a log entry for each deleted row, TRUNCATE TABLE removes all rows in a single operation, making it faster, especially for large tables.

```
TRUNCATE TABLE database.table1;
```

Constraints

Constraints in SQL are rules defined on columns or tables to enforce data integrity and ensure that data values meet certain conditions. Here are some common constraints in SQL:

1. NOT NULL: Ensures that a column cannot contain NULL values.
2. UNIQUE: Ensures that all values in a column are unique.
3. PRIMARY KEY: Combines the NOT NULL and UNIQUE constraints. It uniquely identifies each record in a table.
4. FOREIGN KEY: Ensures referential integrity by enforcing a link between data in two tables.
5. CHECK: Ensures that all values in a column meet specified conditions.
6. DEFAULT: Provides a default value for a column when no value is specified.
7. INDEX: Improves the speed of data retrieval operations on a database table.

Here's how you can define constraints in SQL:

```
CREATE TABLE db.MyTable (

 ID INT PRIMARY KEY,

 Name VARCHAR(50) NOT NULL,

 Age INT CHECK (Age >= 18),

 Email VARCHAR(100) UNIQUE,

 DepartmentID INT);
```

In this example:

- **The `ID` column is the primary key.**
- **The `Name` column cannot contain NULL values.**
- **The `Age` column must be greater than or equal to 18.**
- **The `Email` column must contain unique values.**

Constraints play a vital role in maintaining data integrity and ensuring that databases remain consistent and reliable.

```
    ddl
  Create: db, table
Alter: change
  drop: db, table, column-delete
  Truncate: entries/ rows/ values-delete
```